



CHIJ ST. THERESA'S CONVENT
PRELIMINARY EXAMINATION 2026
SECONDARY 4 EXPRESS / 5 NORMAL (ACADEMIC)

Mark Scheme

CANDIDATE
NAME

CLASS

INDEX
NUMBER

ADDITIONAL MATHEMATICS

4049/2

Paper 2

25 August 2026

2 hours 15 minutes

Candidates answer on the Question Paper.

READ THESE INSTRUCTIONS FIRST

Write your name, class and index number in the spaces at the top of this page.

Write in dark blue or black pen.

You may use a pencil for any diagrams or graphs.

Do not use paper clips, highlighters, glue or correction fluid.

Answer **all** questions.

Give non-exact numerical answers correct to 3 significant figures, or 1 decimal place in the case of angles in degrees, unless a different level of accuracy is specified in the question.

The use of an approved scientific calculator is expected, where appropriate.

You are reminded of the need for clear presentation in your answers.

The number of marks is given in brackets [] at the end of each question or part question.

The total number of marks for this paper is 90.

Mathematical Formula

1. ALGEBRA

Quadratic Equation

For the equation $ax^2 + bx + c = 0$,

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Binomial expansion

$$(a + b)^n = a^n + \binom{n}{1}a^{n-1}b + \binom{n}{2}a^{n-2}b^2 + \dots + \binom{n}{r}a^{n-r}b^r + \dots + b^n,$$

where n is a positive integer and $\binom{n}{r} = \frac{n!}{(n-r)!r!} = \frac{n(n-1)\dots(n-r+1)}{r!}$.

2. TRIGONOMETRY

Identities

$$\sin^2 A + \cos^2 A = 1$$

$$\sec^2 A = 1 + \tan^2 A$$

$$\operatorname{cosec}^2 A = 1 + \cot^2 A$$

$$\sin(A \pm B) = \sin A \cos B \pm \cos A \sin B$$

$$\cos(A \pm B) = \cos A \cos B \mp \sin A \sin B$$

$$\tan(A \pm B) = \frac{\tan A \pm \tan B}{1 \mp \tan A \tan B}$$

$$\sin 2A = 2 \sin A \cos A$$

$$\cos 2A = \cos^2 A - \sin^2 A = 2 \cos^2 A - 1 = 1 - 2 \sin^2 A$$

$$\tan 2A = \frac{2 \tan A}{1 - \tan^2 A}$$

Formulae for $\triangle ABC$

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$

$$a^2 = b^2 + c^2 - 2bc \cos A$$

$$\Delta = \frac{1}{2}ab \sin C$$

- 1 Show that the line $y = 4x$ intersects the curve $y = x^2 + kx + 3 - k$ for all real values of k . [5]

Consider $x^2 + kx + 3 - k = 4x$ [M1]

$$x^2 + (k - 4)x + 3 - k = 0$$

Discriminant, $D = (k - 4)^2 - 4(1)(3 - k)$ [M1]

$$= k^2 - 8k + 16 - 12 + 4k$$

$$= k^2 - 4k + 4$$

$$= (k - 2)^2$$
 [A1]

$D \geq 0$ for all real values of k [M1]

$\Rightarrow$ equation has real roots for all real values of k

$\Rightarrow$ line intersects curve for all real values of k (shown). [AG1]

- 2 The curve $y = x\sqrt{x+1}$ has a stationary value at the point P . Find the coordinates of P . [6]

$$\begin{aligned} \frac{dy}{dx} &= x \frac{d}{dx}(\sqrt{x+1}) + \sqrt{x+1} \frac{d}{dx}(x) \\ &= x \left(\frac{1}{2}(x+1)^{-\frac{1}{2}} \right) + (\sqrt{x+1})(1) \end{aligned}$$
 [M1, M1]

$$= \frac{x}{2\sqrt{x+1}} + \sqrt{x+1}$$

$$= \frac{x + 2(x+1)}{2\sqrt{x+1}}$$
 [M1]

$$= \frac{3x+2}{2\sqrt{x+1}}$$

At the stationary point, $\frac{dy}{dx} = 0$ [M1]

$$\frac{3x+2}{2\sqrt{x+1}} = 0$$

$$x = -\frac{2}{3}$$
 [A1]

When $x = -\frac{2}{3}$, $y = -\frac{2}{3}\sqrt{-\frac{2}{3}+1} = -\frac{2}{3\sqrt{3}} = -\frac{2\sqrt{3}}{9}$ [A1]

The coordinates of P are $\left(-\frac{2}{3}, -\frac{2\sqrt{3}}{9}\right)$.

- 3 (a) The expression $f(x) = ax^3 + bx^2 - 7x + 6$, where a and b are constants, has a factor of $x + 1$ and it leaves a remainder of $\frac{3}{2}$ when divided by $2x - 1$. Find the values of a and b . [4]

$$f(-1) = 0 \Rightarrow a(-1)^3 + b(-1)^2 - 7(-1) + 6 = 0 \quad [\text{M1}]$$

$$\Rightarrow -a + b + 13 = 0 \quad \text{--- (1)}$$

$$f\left(\frac{1}{2}\right) = \frac{3}{2} \Rightarrow a\left(\frac{1}{2}\right)^3 + b\left(\frac{1}{2}\right)^2 - 7\left(\frac{1}{2}\right) + 6 = \frac{3}{2} \quad [\text{M1}]$$

$$\Rightarrow \frac{1}{8}a + \frac{1}{4}b - \frac{7}{2} + 6 = \frac{3}{2}$$

$$\Rightarrow \frac{1}{8}a + \frac{1}{4}b + 1 = 0$$

$$\Rightarrow a + 2b + 8 = 0 \quad \text{--- (2)}$$

$$(1) + (2): \quad 3b + 21 = 0$$

$$b = -7 \quad [\text{A1}]$$

$$\text{Substitute into (1):} \quad a = b + 13 = 6 \quad [\text{A1}]$$

- (b) Solve the equation $4x^3 - 9x^2 + 4 = 0$, expressing the roots in an exact form. [5]

$$\text{Let } f(x) = 4x^3 - 9x^2 + 4.$$

$$f(2) = 4(2)^3 - 9(2)^2 + 4$$

$$= 32 - 36 + 4 = 0$$

$$\text{By the Factor Theorem, } (x - 2) \text{ is a factor of } f(x). \quad [\text{M1}]$$

$$\text{Using long division (or by inspection), the other factor is } 4x^2 - x - 2. \quad [\text{M1}]$$

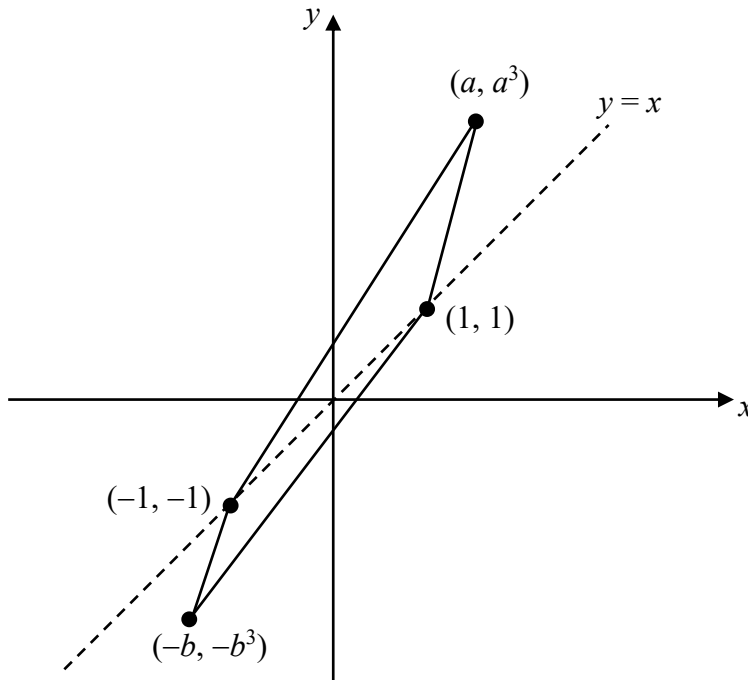
$$x - 2 = 0 \Rightarrow x = 2 \quad [\text{A1}]$$

$$4x^2 - x - 2 = 0 \Rightarrow x = \frac{-(-1) \pm \sqrt{(-1)^2 - 4(4)(-2)}}{2(4)} \quad [\text{M1}]$$

$$= \frac{1 \pm \sqrt{33}}{8} \quad [\text{A1}]$$

$$\therefore x = 2 \text{ or } x = \frac{1 \pm \sqrt{33}}{8}$$

- 4 A quadrilateral is formed by the four points (a, a^3) , $(-b, -b^3)$, $(1, 1)$ and $(-1, -1)$, where the constants a and b are such that $a > b > 1$. Express the area of the quadrilateral as a product of two factors, one of which is $(a + b)$. [4]



$$\text{Area of quadrilateral} = \frac{1}{2} \begin{vmatrix} 1 & a & -1 & -b & 1 \\ 1 & a^3 & -1 & -b^3 & 1 \end{vmatrix} \quad [\text{M1}]$$

$$= \frac{1}{2} [a^3 - a + b^3 - b - (-b^3 + b - a^3 + a)] \quad [\text{M1}]$$

$$= \frac{1}{2} [2a^3 - 2a + 2b^3 - 2b]$$

$$= a^3 + b^3 - (a + b) \quad [\text{A1}]$$

$$= (a + b)(a^2 - ab + b^2) - (a + b)$$

$$= (a + b)(a^2 - ab + b^2 - 1) \quad [\text{A1}]$$

5 Solve the equation $\lg x^3 = 1 - \log_x(0.0001)$. [6]

$$3\lg x = 1 - \log_x(0.0001) \quad [\text{M1}]$$

$$= 1 - \frac{\lg(0.0001)}{\lg x} \quad [\text{M1}]$$

$$= 1 - \frac{(-4)}{\lg x} \quad [\text{A1}]$$

$$= 1 + \frac{4}{\lg x}$$

Let $u = \lg x$.

Then $3u = 1 + \frac{4}{u}$

$$3u^2 = u + 4 \quad [\text{M1} - \text{quadratic}]$$

$$3u^2 - u - 4 = 0$$

$$(3u - 4)(u + 1) = 0$$

$$u = \frac{4}{3} \quad \text{or} \quad u = -1 \quad [\text{A1}]$$

$$\lg x = -\frac{2}{3} \quad \text{or} \quad \lg x = -1$$

$$x = 10^{-\frac{2}{3}} \quad \text{or} \quad x = 10^{-1}$$

$$x = 0.215 \text{ (3 s.f.)} \quad \text{or} \quad x = 0.1 \quad [\text{A1}]$$

6 The equation of a curve is $y = e^{-\frac{1}{2}x} \cos x$.

(i) Find $\frac{dy}{dx}$. [2]

$$\frac{dy}{dx} = \frac{d}{dx} \left(e^{-\frac{1}{2}x} \right) \cos x + e^{-\frac{1}{2}x} \frac{d}{dx} (\cos x) \quad [\text{M1}]$$

$$= \left(-\frac{1}{2} e^{-\frac{1}{2}x} \right) \cos x + e^{-\frac{1}{2}x} \frac{d}{dx} (-\sin x)$$

$$= -e^{-\frac{1}{2}x} \left(\frac{1}{2} \cos x + \sin x \right) \quad [\text{A1}]$$

The curve intersects the y -axis at the point A .

(ii) Find the gradient of the normal to the curve at the point. [2]

$$\text{At the } y\text{-axis, } x = 0 \text{ and } \frac{dy}{dx} = -e^{-\frac{1}{2}(0)} \left(\frac{1}{2} \cos 0 + \sin 0 \right)$$

$$\text{gradient of tangent} = -\frac{1}{2} \quad [\text{B1}]$$

$$\text{gradient of normal} = -1 / \left(-\frac{1}{2} \right) = 2 \quad [\text{B1}]$$

The smallest positive value of x at which the curve has a stationary point is $x = x_0$.

(iii) Find the value of x_0 . [3]

$$\text{At the stationary points, } \frac{dy}{dx} = 0 \quad [\text{M1}]$$

$$-e^{-\frac{1}{2}x} \left(\frac{1}{2} \cos x + \sin x \right) = 0$$

$$\frac{1}{2} \cos x + \sin x = 0 \quad \text{since } e^{-\frac{1}{2}x} > 0 \text{ for all } x \text{ real}$$

$$\sin x = -\frac{1}{2} \cos x$$

$$\tan x = -\frac{1}{2} \quad [\text{M1}]$$

$$\text{Basic angle, } \alpha = \tan^{-1} \left(\frac{1}{2} \right) = 0.46365 \text{ (5 s.f.)}$$

$$\text{The first positive value of } x \text{ is } x_0 = \pi - \alpha = 2.68 \text{ rad (3 s.f.)} \quad [\text{A1}]$$

- 7 (a) Prove the identity $\tan\left(x + \frac{\pi}{4}\right) = \frac{1 + \sin 2x}{\cos 2x}$. [4]

$$\begin{aligned}
 \text{LHS} &= \frac{\tan x + \tan \frac{\pi}{4}}{1 - \tan x \tan \frac{\pi}{4}} && [\text{M1}] \\
 &= \frac{\tan x + 1}{1 - \tan x} \\
 &= \frac{\frac{\sin x}{\cos x} + 1}{1 - \frac{\sin x}{\cos x}} \\
 &= \frac{\sin x + \cos x}{\cos x - \sin x} && [\text{M1}] \\
 &= \frac{\cos x + \sin x}{\cos x - \sin x} \times \frac{\cos x + \sin x}{\cos x + \sin x} && [\text{M1}] \\
 &= \frac{\cos^2 x + \sin^2 x + 2 \cos x \sin x}{\cos^2 x - \sin^2 x} \\
 &= \frac{1 + \sin 2x}{\cos 2x} && [\text{AG1}] \\
 &= \text{RHS (proved)}
 \end{aligned}$$

(b) Hence solve the equation $1 + \sin 2x = 3 \cos 2x$ for $-2\pi \leq x \leq 2\pi$. [6]

$$\frac{1 + \sin 2x}{\cos 2x} = 3$$

$$\tan\left(x + \frac{\pi}{4}\right) = 3 \quad [\text{M1}]$$

Basic angle, $\alpha = \tan^{-1}(3) = 1.2490 \text{ rad}$ [A1]

Since $\tan\left(x + \frac{\pi}{4}\right) = 3 (> 0)$, $x + \frac{\pi}{4}$ is in the 1st or 3rd quadrant.

$$-2\pi \leq x \leq 2\pi \quad \Rightarrow \quad -\frac{7\pi}{4} \leq x \leq \frac{9\pi}{4}$$

$$x + \frac{\pi}{4} = \alpha, \quad \pi + \alpha, \quad -(\pi - \alpha), \quad -(2\pi - \alpha) \quad [\text{M2}]$$

$$x = 0.464, \quad 3.61, \quad -2.68, \quad -5.82 \quad [\text{A2}]$$

8 The equation of a curve is $y = \sec x - \tan x$.

(i) Show that $\frac{dy}{dx} = -\left(\frac{1}{1+\sin x}\right)$. [5]

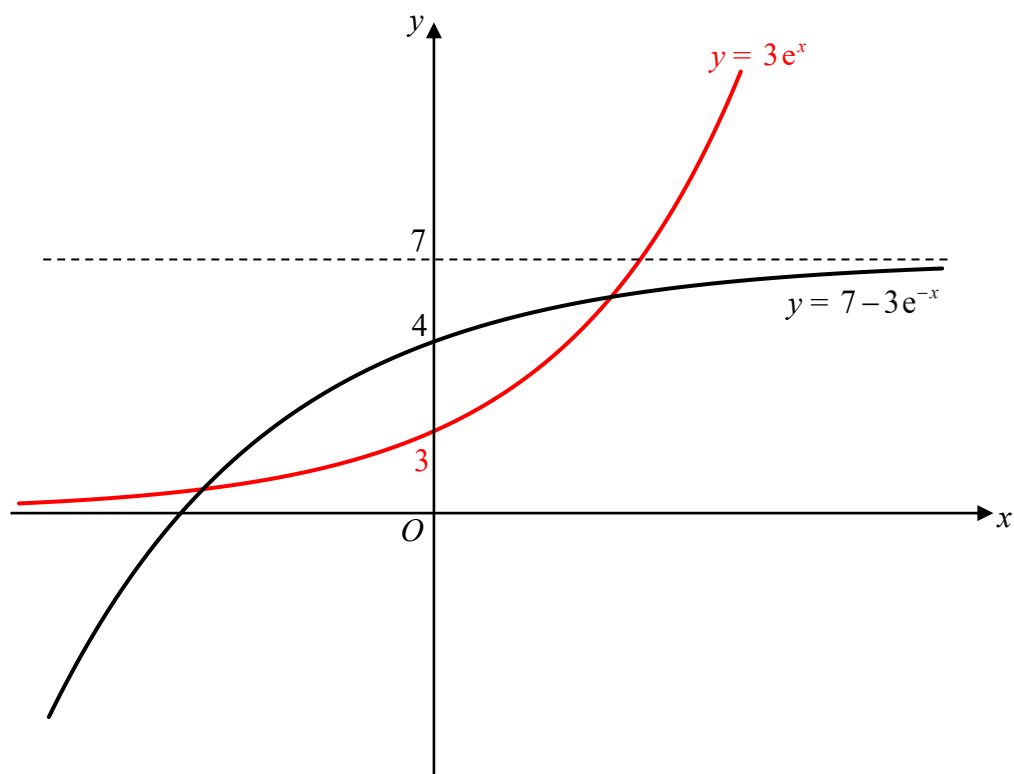
$$\begin{aligned}
 y &= \frac{1}{\cos x} - \tan x \\
 \frac{dy}{dx} &= \frac{\cos x \frac{d}{dx}(1) - (1) \frac{d}{dx}(\cos x)}{\cos^2 x} - \frac{d}{dx}(\tan x) \\
 &= \frac{-(1) \frac{d}{dx}(-\sin x)}{\cos^2 x} - \frac{d}{dx}(\tan x) && \text{[M1, M1]} \\
 &= \frac{\sin x}{\cos^2 x} - \sec^2 x && \text{[M1]} \\
 &= \frac{\sin x}{\cos^2 x} - \frac{1}{\cos^2 x} \\
 &= \frac{\sin x - 1}{\cos^2 x} \\
 &= \frac{\sin x - 1}{1 - \sin^2 x} \\
 &= \frac{\sin x - 1}{(1 - \sin x)(1 + \sin x)} && \text{[M1]} \\
 &= -\left(\frac{1}{1 + \sin x}\right) && \text{[AG1]}
 \end{aligned}$$

The line L is a tangent to the curve at the point where $x = -\frac{\pi}{6}$.

(ii) Find the y -intercept of L . [3]

$$\begin{aligned}
 \text{When } x = -\frac{\pi}{6}, \quad y &= \sec\left(-\frac{\pi}{6}\right) - \tan\left(-\frac{\pi}{6}\right) \\
 &= \frac{2}{\sqrt{3}} - \left(-\frac{1}{\sqrt{3}}\right) = \sqrt{3} && \text{[B1 – accept 3 s.f.]} \\
 \text{and} \quad \frac{dy}{dx} &= -\left[\frac{1}{1 + \sin\left(-\frac{\pi}{6}\right)}\right] = -2 && \text{[B1]} \\
 L: \quad y - \sqrt{3} &= -2\left[x - \left(-\frac{\pi}{6}\right)\right] \\
 y &= -2x - \frac{\pi}{3} + \sqrt{3} \\
 \text{The } y\text{-intercept is } \sqrt{3} - \frac{\pi}{3} \quad (&= 0.685 \text{ correct to 3 s.f.}) && \text{[B1]}
 \end{aligned}$$

- 9 The diagram shows the curve of $y = 7 - 3e^{-x}$.



- (i) Draw the graph of $y = 3e^x$ on the same axes, indicating clearly the axial intercepts, if any. [2]
- (ii) Solve the equation $7 - 3e^{-x} = 3e^x$. [4]

$$7e^x - 3 = 3(e^x)^2$$

Let $u = e^x$.

Then

$$7u - 3 = 3u^2$$

[M1 – quadratic]

$$3u^2 - 7u + 3 = 0$$

$$u = \frac{-(-7) \pm \sqrt{(-7)^2 - 4(3)(3)}}{2(3)} \quad \text{[M1]}$$

$$e^x = \frac{7 \pm \sqrt{13}}{6} \quad \text{[A1]}$$

$$x = \ln\left(\frac{7 \pm \sqrt{13}}{6}\right) \quad (= 0.570 \text{ or } -0.570 \text{ to 3 s.f.}) \quad \text{[A1]}$$

10 The equation of the upper half of a circle is $y = 4 + \sqrt{16 + 6x - x^2}$.

- (i) Find $\frac{dy}{dx}$, giving your answer in the form $\frac{a+bx}{\sqrt{16+6x-x^2}}$, where a and b are constants to be found. [2]

$$\begin{aligned}\frac{dy}{dx} &= \frac{d}{dx}(4) + \frac{d}{dx}\sqrt{16+6x-x^2} \\ &= 0 + \frac{1}{2}(16+6x-x^2)^{-\frac{1}{2}}(6-2x) & [M1] \\ &= \frac{3-x}{\sqrt{16+6x-x^2}} & [A1]\end{aligned}$$

- (ii) Show that the radius of the circle is 5 units and find the coordinates of C , the centre of the circle. [3]

$$\begin{aligned}y - 4 &= \sqrt{16 + 6x - x^2} \\ (y - 4)^2 &= 16 + 6x - x^2 \\ x^2 - 6x + (y - 4)^2 &= 16 & [M1] \\ (x - 3)^2 - (3)^2 + (y - 4)^2 &= 16 & [M1] \\ (x - 3)^2 + (y - 4)^2 &= 25 \\ \text{Radius of circle is 5 units (shown) and coordinates of } C &\text{ are } (3, 4). & [A1]\end{aligned}$$

The points P and Q on the circle are such that the tangents to the circle at these two points both have a gradient of $\frac{3}{4}$.

(iii) Find the equations of these two tangents.

[6]

$$\frac{dy}{dx} = \frac{3}{4}$$

$$\frac{3-x}{\sqrt{16+6x-x^2}} = \frac{3}{4}$$

$$4(3-x) = 3\sqrt{16+6x-x^2}$$

$$4^2(3-x)^2 = 3^2(16+6x-x^2) \quad [\text{M1} - \text{quadratic}]$$

$$16(9-6x+x^2) = 9(16+6x-x^2)$$

$$144-96x+16x^2 = 144+54x-9x^2$$

$$25x^2-150x = 0$$

$$x(x-6) = 0$$

$$x = 0 \text{ or } x = 6 \quad [\text{A1}]$$

$$\text{When } x = 0, y = 4 + \sqrt{16+6(0)-(0)^2} = 8 \quad [\text{M1}]$$

Let this be point P .

$$\text{Equation of tangent at } P \text{ is } y-8 = \frac{3}{4}(x-0)$$

$$y = \frac{3}{4}x + 8 \quad [\text{A1}]$$

Since the tangents at P and Q are parallel, PQ is a diameter of the circle with P in the upper half of the circle and Q in the lower half.

To find the coordinates of Q , we substitute $x = 6$ into the equation of the full circle.

$$(6-3)^2 + (y-4)^2 = 25$$

$$(y-4)^2 = 25-9 = 16$$

$$y-4 = 4 \text{ or } -4$$

$$y = 8 \text{ (upper half, reject) or } y = 0 \text{ (lower half)} \quad [\text{M1}]$$

$$\text{Equation of tangent at } Q \text{ is } y-0 = \frac{3}{4}(x-6)$$

$$y = \frac{3}{4}x - \frac{9}{2} \quad [\text{A1}]$$

- 11 The temperature, $T^{\circ}\text{C}$, of the coffee in a cup t minutes after boiling water is poured into the cup is given by $T = 28 + Ae^{-kt}$, where A and k are constants.

The table below shows corresponding values of t and T .

t	5	10	15	20	25
T	87.0	76.2	67.6	60.3	54.5
$\ln(T - 28)$	4.08	3.88	3.68	3.48	3.28

- (i) Plot $\ln(T - 28)$ against t and draw a straight line graph to illustrate the information. [2]

- (ii) Use your graph to estimate the values of the constants A and k . [4]

$$T = 28 + Ae^{-kt}$$

$$T - 28 = Ae^{-kt}$$

$$\ln(T - 28) = \ln A + \ln e^{-kt} \quad [\text{M1}]$$

$$= (-k)t + (\ln A) \quad [\text{M1}]$$

Gradient of the line is $(-k)$ and the vertical intercept is $(\ln A)$.

From the graph, vertical intercept is $(\ln A) = 4.28 \Rightarrow A = 72.2$ [A1]

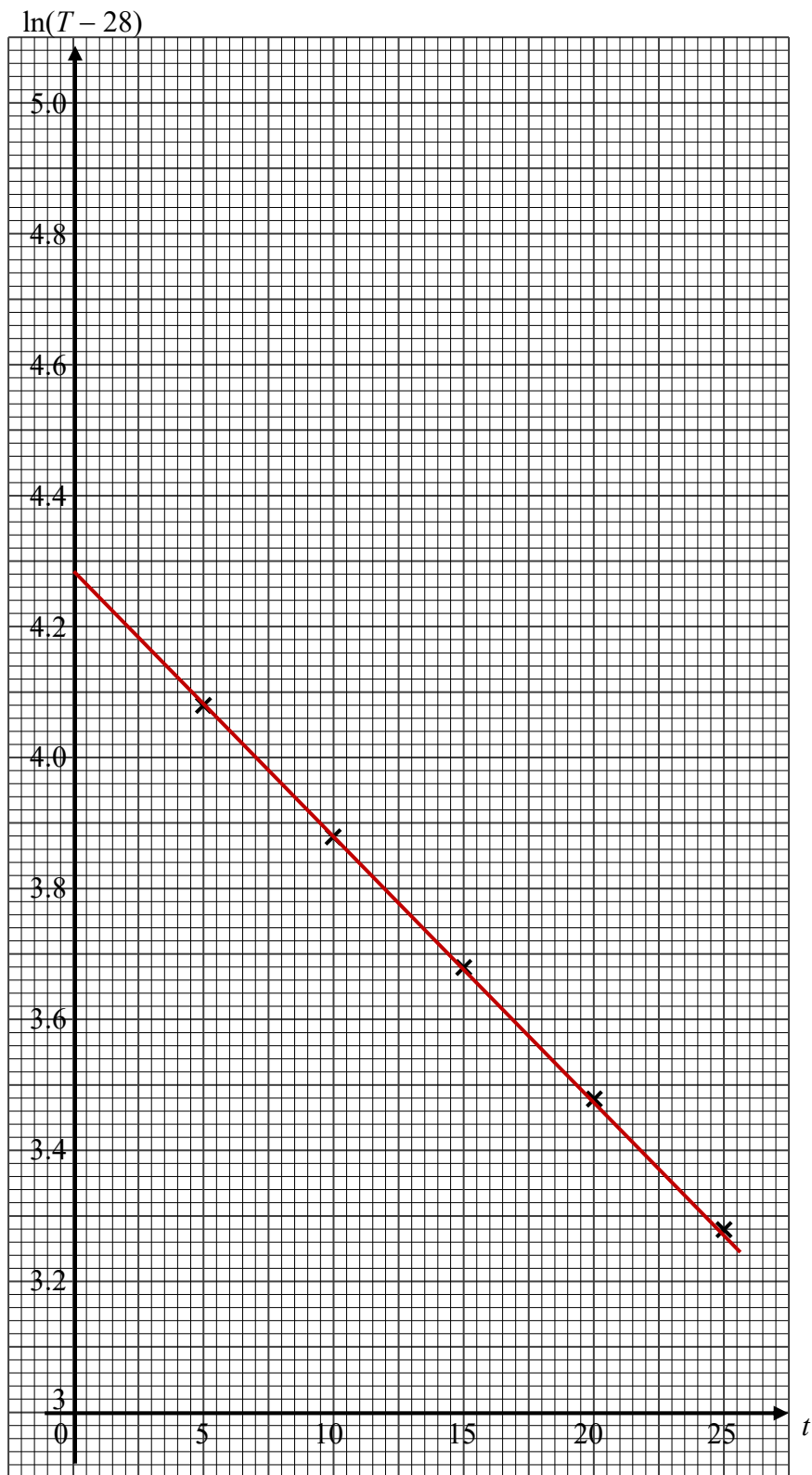
Gradient is $(-k) = \frac{3.28 - 4.28}{25 - 0} = -0.04$ so $k = 0.04$ [A1]

- (iii) Use your graph to estimate how long it takes for the temperature of the coffee to drop to 70°C . [2]

When $T = 70$, $\ln(T - 28) = 3.74$

From the graph, this corresponds to $t = 13.5$. [M1]

It takes about 13.5 minutes for the temperature to drop to 70°C . [A1]



- 12 A particle travels in a straight line and it is initially at the point O . The displacement, s metres from O , of the particle at time t seconds is given $s = te^{-0.1t}$.

(i) Obtain an expression for the velocity of the particle at time t seconds. [2]

$$\begin{aligned} \text{Velocity, } v &= \frac{ds}{dt} = \frac{d}{dt}(t)(e^{-0.1t}) + t \frac{d}{dt}(e^{-0.1t}) & [\text{M1}] \\ &= (1)(e^{-0.1t}) + t(-0.1e^{-0.1t}) \\ &= (1 - 0.1t)e^{-0.1t} & [\text{A1}] \end{aligned}$$

(ii) Determine when the particle comes to instantaneous rest. [2]

$$\begin{aligned} \text{Particle comes to instantaneous rest when } v &= 0 \\ (1 - 0.1t)e^{-0.1t} &= 0 & [\text{M1}] \\ 1 - 0.1t &= 0 \\ t &= 10 & [\text{A1}] \end{aligned}$$

(iii) Find the distance travelled by the particle in the first 12 seconds. [2]

$$\begin{aligned} \text{When } t = 0, \quad s &= 0 \\ \text{When } t = 10, \quad s &= 10e^{-1} \\ \text{When } t = 12, \quad s &= 12e^{-1.2} \\ \text{Distance travelled in first 12 seconds is } &(10e^{-1} - 0) + (10e^{-1} - 12e^{-1.2}) & [\text{M1}] \\ &= 3.74 \text{ m} & [\text{A1}] \end{aligned}$$

(iv) Show that the particle does not accelerate in the first 20 seconds.

[4]

$$\text{Acceleration, } a = \frac{dv}{dt} = e^{-0.1t} \frac{d}{dt}(1 - 0.1t) + (1 - 0.1t) \frac{d}{dt}(e^{-0.1t}) \quad [\text{M1}]$$

$$= e^{-0.1t}(-0.1) + (1 - 0.1t)(-0.1e^{-0.1t})$$

$$= e^{-0.1t}[(-0.1) + (1 - 0.1t)(-0.1)]$$

$$= 0.1(0.1t - 2)e^{-0.1t} \quad [\text{A1}]$$

When $t \leq 20$, $(0.1t - 2) \leq 0$.

Since $0.1e^{-0.1t} > 0$ for all t values,

$$a = 0.1(0.1t - 2)e^{-0.1t} \leq 0 \quad \text{for } t \leq 20 \quad [\text{M1}]$$

Hence the particle does not accelerate in the first 20 seconds (shown). [AG1]

~ ~ ~ End of Paper ~ ~ ~